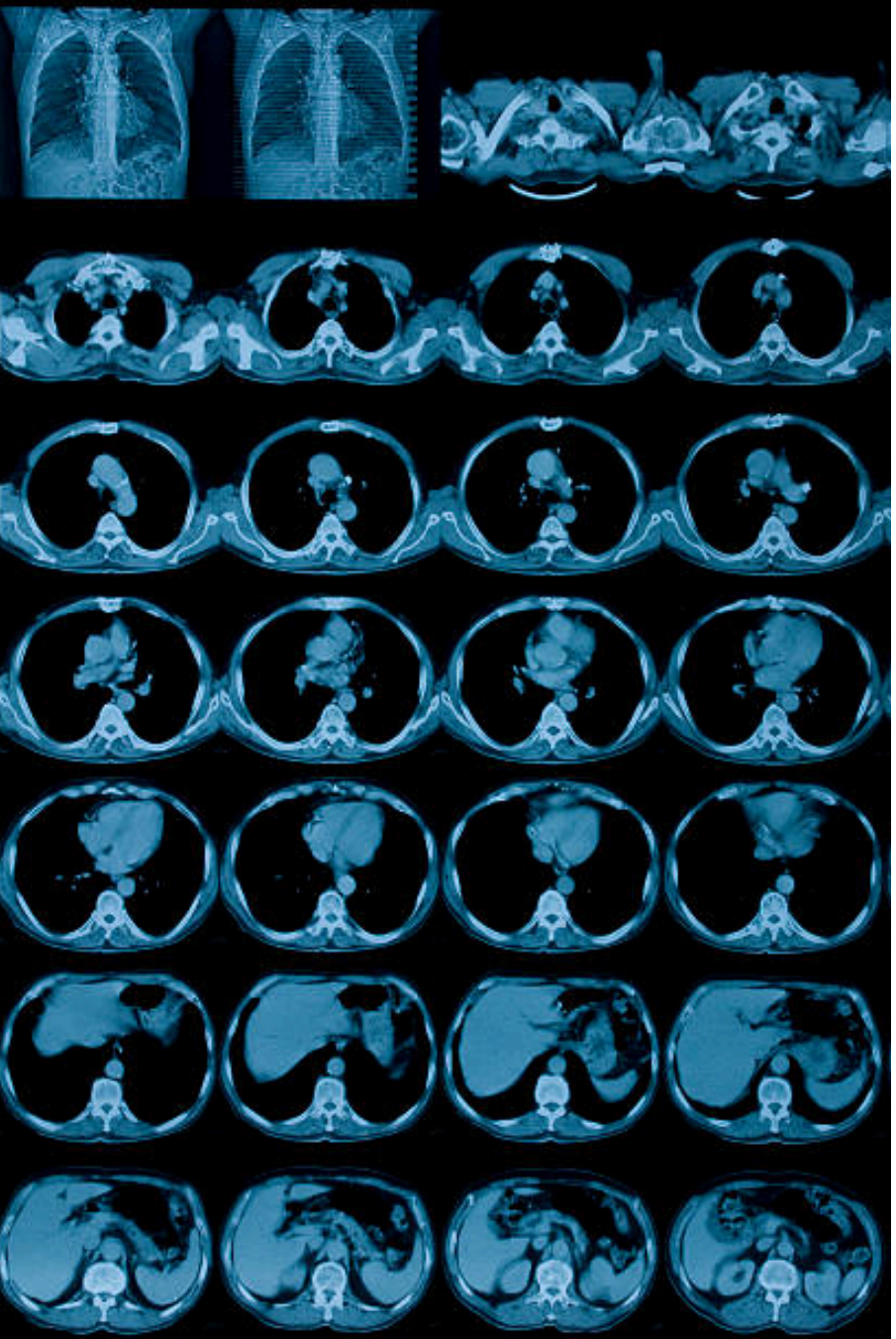


Volumetric CT Scan Analysis for Pulmonary Nodule Detection and Malignancy Classification Using 3D CNN Architectures

Nodule AI

Guide: J. Ravindra Babu, Assistant Professor
CSE-Data Science

Team-14

B. Sandeep Raghavendra (22JR1A4439)
A. Jaswanth Kumar (22JR1A4434)
G. Vignan (22JR1A4448)
J. Ganesh (22JR1A4452)
K. Madhu (22JR1A4457)

Introduction: The Challenge of Early Detection

Leading Cause

Lung cancer remains one of the leading causes of cancer-related mortality worldwide, with millions affected annually.

Early Detection Matters

Early detection of pulmonary nodules significantly increases survival rates and improves patient outcomes dramatically.

CT Scans Standard

Computed Tomography (CT) scans are the standard imaging modality for screening, providing detailed volumetric data.

AI Solution

Automated, AI-driven analysis assists radiologists in detecting and classifying lung nodules with precision.

Problem Statement

Data Imbalance

Classimbalanceinmedical datasets often leads to biased AI models that perform poorly on rare malignant cases.

Computational Cost

Processingfull 3Disotropic medical volumes demands enormous computational resources and time.

Hardware Limitations

Legacydatasets causesevere hardware exhaustion during local training, limiting accessibility.

Visualization Gap

Lackofinteractive,explainable, and visually intuitive 3D visualization for clinicians.

Existing Systems vs. Limitations

Traditional CAD Systems

- Rely heavily on hand-crafted features (shape, texture, intensity)
- High false-positive rates
- Brittle generalization to new data

Standard 3D CNNs

- Massive GPU memory requirements
- Difficult deployment in resource-constrained environments

Manual Review Challenges

- Slice-by-slice 2D inspection of 3D scans
- Cognitively taxing for radiologists
- Prone to human error and fatigue

Critical Gap: Current solutions either compromise accuracy or require prohibitively expensive hardware.

Proposed Solution: Nodule AI

Dataset Migration

MigratedtobalancedLUNA-22 (ISMI) 3D patches, solving class imbalance issues.

Lightweight Model

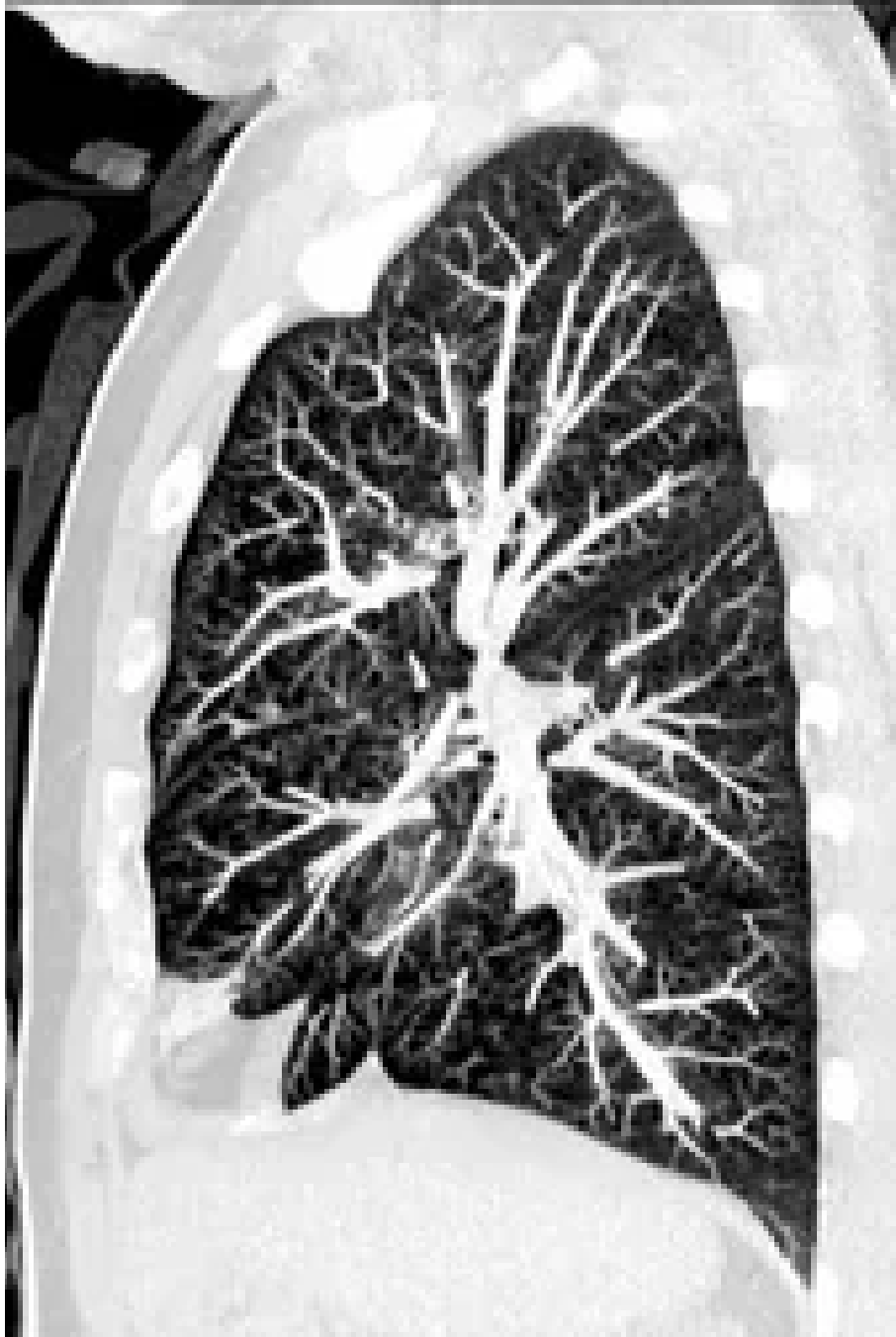
DeeplyoptimizedU-Net 3D with Depthwise Separable Convolutions.

Minimal Footprint

155Kparameters(~1MB) preserving spatial context.

Web Application

Real-time3Drendering with clinical reasoning interface.



System Objectives

Real-Time Classification
Highly accurate 3D lung nodule classification (Benign vs. Malignant) with minimal latency.

Edge-Ready Architecture
Lightweight enough to run inference on edge devices and CPUs without dedicated GPUs.

Visual Excellence
Visually stunning, CT-authentic 3D interface designed specifically for radiologists.

Clinical Descriptions
Rich clinical reports detailing nodule size, texture, and recommended actions.

Data Collection & Preprocessing

LUNA-22 Dataset

1,176 pre-extracted 3D patches in Medical '.nii.gz' volumetric arrays format.

Preprocessing Pipeline

Voxel intensity normalization to [0,1] scale1.

Spatial resampling with trilinear interpolation2.

Standardized dimensions (64×64×32)3.

Automated cloud data-discovery4.

Development Process & Feasibility

1

Phase 1
Environment setup and legacy code purge

2

Phase 2
Neural architecture redesign

3

Phase 3
API refactoring and optimization

4

Phase 4
Cloud training pipeline generation

5

Phase 5
UI/UX and 3D visualizer enhancements

Technical
Robust stacks: PyTorch, Flask, Next.js, Three.js eliminate enterprise GPU requirements.

Operational
Web-browser deployment requires zero installation for clinicians.

Economic
Free tier cloud environments for training; local inference practically free.

Development Process & Feasibility

1. **Phase 1**
   Environment setup and legacy code purge
2. **Phase 2**
   Neural architecture redesign
3. **Phase 3**
   API refactoring and optimization
4. **Phase 4**
   Cloud training pipeline generation
5. **Phase 5**
   UI/UX and 3D visualizer enhancements

Technical

Robust stacks: PyTorch, Flask, Next.js, Three.js eliminate enterprise GPU requirements.

Operational

Web-browser deployment requires zero installation for clinicians.

Economic

Free tier cloud environments for training; local inference practically free.

Lightweight U-Net 3D Architecture

Why U-Net?

U-shape architecture excels in medical imaging by concatenating deep semantic features with shallow spatial features via skip connections.

Why Lightweight?

Standard 3D convolutions require massive FLOPs. Depthwise Separable 3D Convolutions cut computations drastically.

Model Size

Final model: ~0.59 MB (154,812 parameters) enabling CPU inference.

Training Strategy

- **Loss Function:** Combined BCEWithLogitsLoss + Soft Dice Loss
- **Optimizer:** AdamW with Cosine Annealing scheduler
- **Bias Trick:** Final layer bias (-5.0) prevents explosive gradients
- **Hardware:** Automated Mixed Precision (AMP) on Kaggle GPUs

Impact & Expected Results

155K	0.59MB	64×64×32
Model Parameters		
Ultra-lightweight architecture	Minimal deployment size	Consistent preprocessing

Applications

- Triage systems in overcrowded departments
- Second-reader concurrent review
- Medical education tool
- Rural health deployment

Expected Outcomes

- Lower screening turnaround time
- Reduced false-negative rates
- Democratized AI access via browser
- Enhanced patient outcomes

Conclusion

Nodule AI bridges complex 3D deep learning with practical clinical deployment. By optimizing a Lightweight U-Net 3D embedded in a visually rich, zero-install web application, we've created a robust tool for pulmonary analysis that is scalable, fast, and represents a significant step towards intelligent, automated medical diagnosis.

Thank you